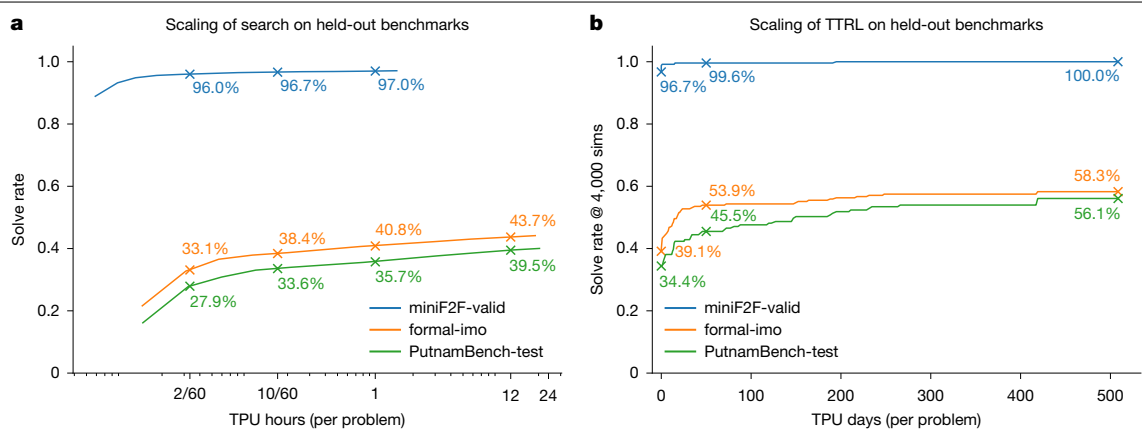

a
Scaling of search on held-out benchmarks
Solve rate
1.0
0.8
0.6
0.4
0.2
0
96.0%
96.7%
97.0%
33.1%
38.4%
40.8%
43.7%
27.9%
33.6%
35.7%
39.5%
miniF2F-valid
formal-imo
PutnamBench-test
2/60
10/60
1
12
24
TPU hours (per problem)
b
Scaling of TTRL on held-out benchmarks
Solve rate @ 4,000 sims
1.0
0.8
0.6
0.4
0.2
0
96.7%
99.6%
100.0%
53.9%
58.3%
45.5%
56.1%
39.1%
34.4%
miniF2F-valid
formal-imo
PutnamBench-test
0
100
200
300
400
500
TPU days (per problem)